

EDUCATION

UNIVERSITY OF COLORADO BOULDER

2019
MS in Aerospace Engineering
MS Thesis: numerical optimization

CORNELL UNIVERSITY

2018
BS in Electrical and Computer Engineering
Minor: Aerospace Eng. & Archaeology

SOFTWARE

Proficient

C/C++ • Python • Git
Matlab • $\LaTeX$

Design:

Fusion360 • EAGLE

HARDWARE

Circuit Design and PCB Layout
Embedded Systems Development
Mechanical Design
3D Printing
Machining

AWARDS

2018: Matthew Isakowitz Fellow
2017: Winner Caltech Space Challenge
MakeMIT Amazon Prize

INDUSTRY

SPACEX | 2020 - PRESENT**GUIDANCE, NAVIGATION, AND CONTROL ENGINEER II****STARSHIP AND LUNAR STARSHIP**

- Responsible Engineer for Starship Flight 9
- Contributed across all Starship integrated flight tests: Monte Carlo simulation, algorithm development, propulsion modeling, configuration automation, and flip/landing burn design
- Responsible Engineer for Lunar Landing: designed and implemented the full guidance, control, and allocation stack from cislunar space to the lunar surface; helped architect modular simulation and monte carlo infrastructure to support rapid feature development
- Built onboard real-time convex optimization routines for rocket landing using scratch-built, robust solvers
- Developed offline sequential convex programming tools for mission and trajectory design
- Led GNC on \$150M+ in NASA milestone products
- Developed manual landing control schema with NASA Astronaut Office; led technical demonstrations and maintained ongoing stakeholder relationships

STARSHIELD

- Cradle-to-grave GNC development for a new program: wrote and flew 12+ novel algorithms; demonstrated successful fault detection, recovery, on-orbit operations, and completed full data review cycle on a team of four
- Designed and implemented attitude control, orbit maintenance and maneuvering, state machine logic, device deployment sequences, and FDIR for peripheral systems
- Sole maintainer of orbit maneuvering and collision avoidance for a new satellite constellation; a dedicated successor team was stood up upon transition

SPACEX | INTERNSHIP/ASSOCIATE POSITIONS | SUMMERS 2015/16/18/19

- Four internships across three sites spanning Falcon, Dragon, and Starlink
- Pad 39A launch instrumentation; Dragon RF reradiation system for TDRS checkouts; Starlink ADCS avionics; Gateway antenna actuation circuit design

CORNELL SPACE SYSTEM DESIGN STUDIO AUG 2014 – NOV 2017 | ITHACA, NY

- Avionics Lead on Artificial Gravity Cubesat under Dr Daniel Selva; complete in-house avionics design to demonstrate controlled artificial gravity.
- Avionics Lead on the Violet Nanosatellite Project under Dr. Mason Peck.

PERSONAL PROJECTS

THE NOTFORFLIGHT LANDER PROJECT | '20 - PRESENT | WWW.NOTFORFLIGHT.COM

- Pressure-fed liquid bipropellant VTVL testbed for novel guidance and control algorithm development
- Fully scratch-built: film-cooled liquid rocket engine, electric TVC actuators, throttle valve, custom avionics, and a complete C++ flight software and GNC stack
- Fluid systems and flight software under active test; targeting static fire summer 2026

PUBLICATIONS

JAN '21 – P. Lysandrou, RD Braun, A 6-DoF Successive Convexification Powered Descent Guidance Implementation using Modified Rodrigues Parameters, AIAA Scitech 2021 Forum (Conference)